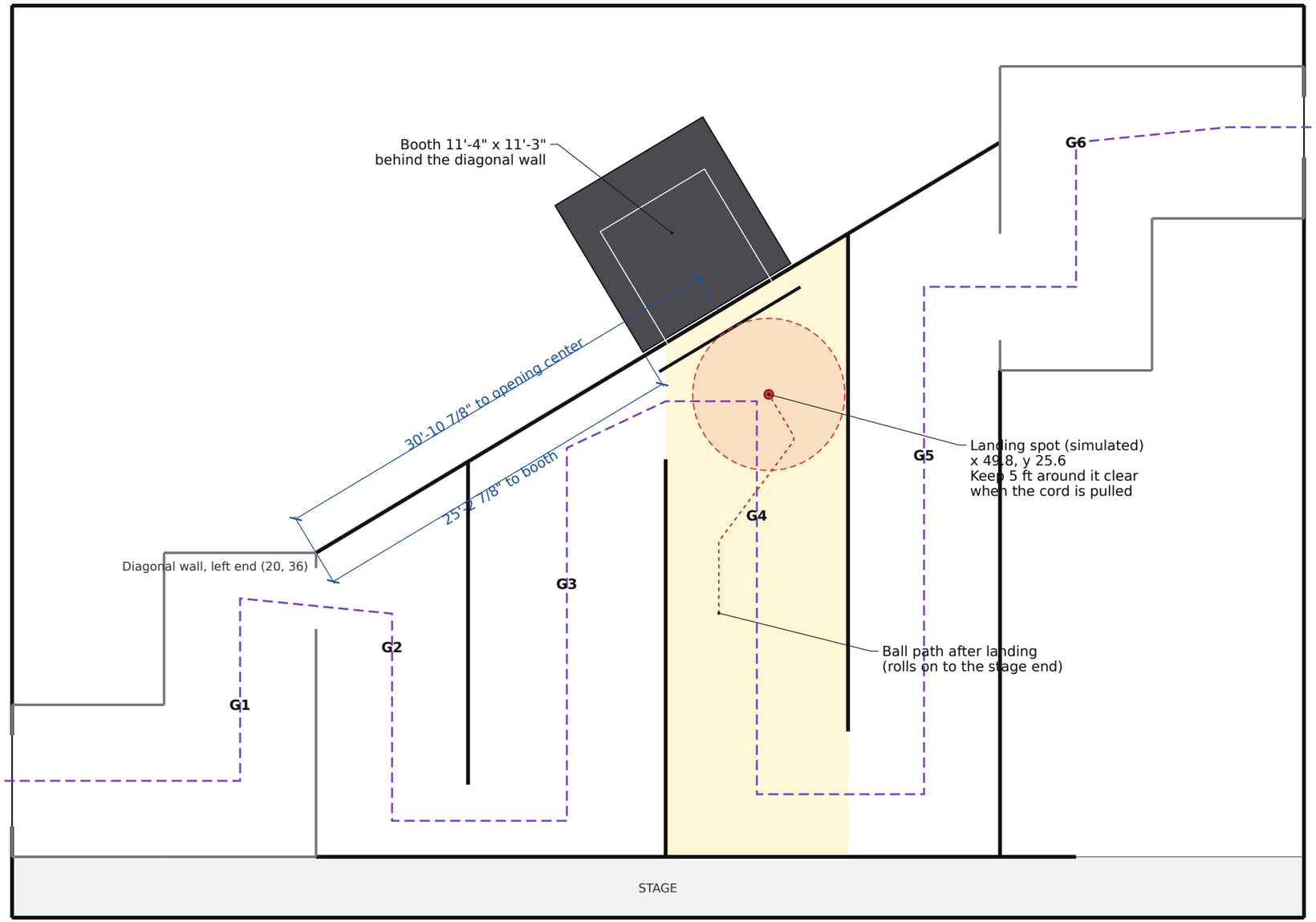


SITE PLAN: GYM, 85 x 60 FT

Plan coordinates: x from the left wall, y from the top wall (stage along the bottom). Group 4 is the third lane.

WHAT IS BEING BUILT

A booth behind the diagonal wall holds the 6.5 ft ball on a 12 degree ramp. One pull on a cord drops a black kabuki drape and frees a hinged stop bar; the ball rolls over the 8 ft wall and drops into the front of the group 4 lane, then rolls down the lane after the group and stops at the stage end.

KEY NUMBERS

Item	Value
Opening center on the wall	30'-10 7/8" from left end
Opening (clear)	8'-0" wide x to 16'-3"
Booth (outside)	11'-4" W x 11'-3" D x 17'-0"
Ramp	8'-0" wide, 12 deg, 9'-11 5/8" long
Deck at lip / lane face	8'-1" / 8'-3 5/8"
Deck at high end	10'-1 7/8"
Ball center at rest	12'-9 1/8" high
Kabuki drape	10'-10" x 8'-9", hem 7'-9"
Operator platform	4'-0" x 2'-5 1/2", top 5'-0"
Ball leaves lip	2.6 s after the pull
Ball lands	3.5 s, 6'-1 5/8" out from wall

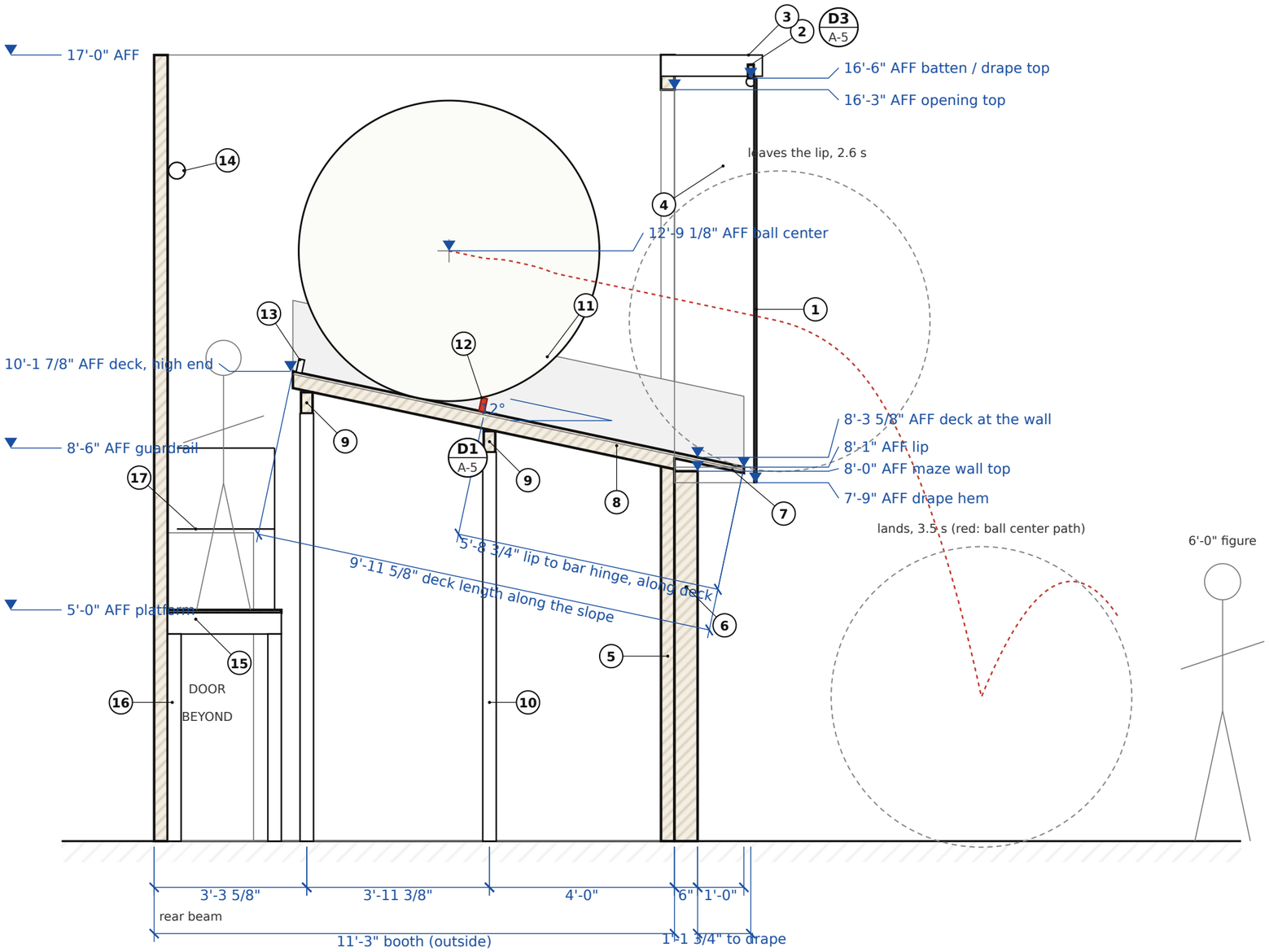
SHEETS

A-1 Site plan and key numbers
A-2 Section A-A through the ramp
A-3 Elevations from the lane
A-4 Plan of the rig
A-5 Mechanism details: stop bar and kabuki drop
A-6 Rigging and the release sequence
A-7 Framing, cut list, and hardware
A-8 Operating, reset, and safety

Conventions

AFF = above finished floor (the gym floor).
"Out" is measured square to the diagonal wall, toward the lane, from the wall centerline; the maze wall is 6 in thick, 8 ft tall.
Lumber is nominal (2x4 = 1 1/2 x 3 1/2 in actual).

GYM HAUNTED HOUSE: HIDDEN BALL DROP (GROUP 4)	SHEET
Site plan and key numbers Scale: 1/8 in = 1 ft (site plan) (on a 17 x 11 in sheet) Planning sketch: verify every dimension on site before cutting. From the sandbox model (src/scenes/ball-drop-rig.js), scripts/drawings.mjs	**A-1** 2026-10-17



SECTION A-A

Cut through the middle of the opening, square to the wall, looking toward the wall's right-hand end. Booth left, lane right. Ball at rest on the stop bar.

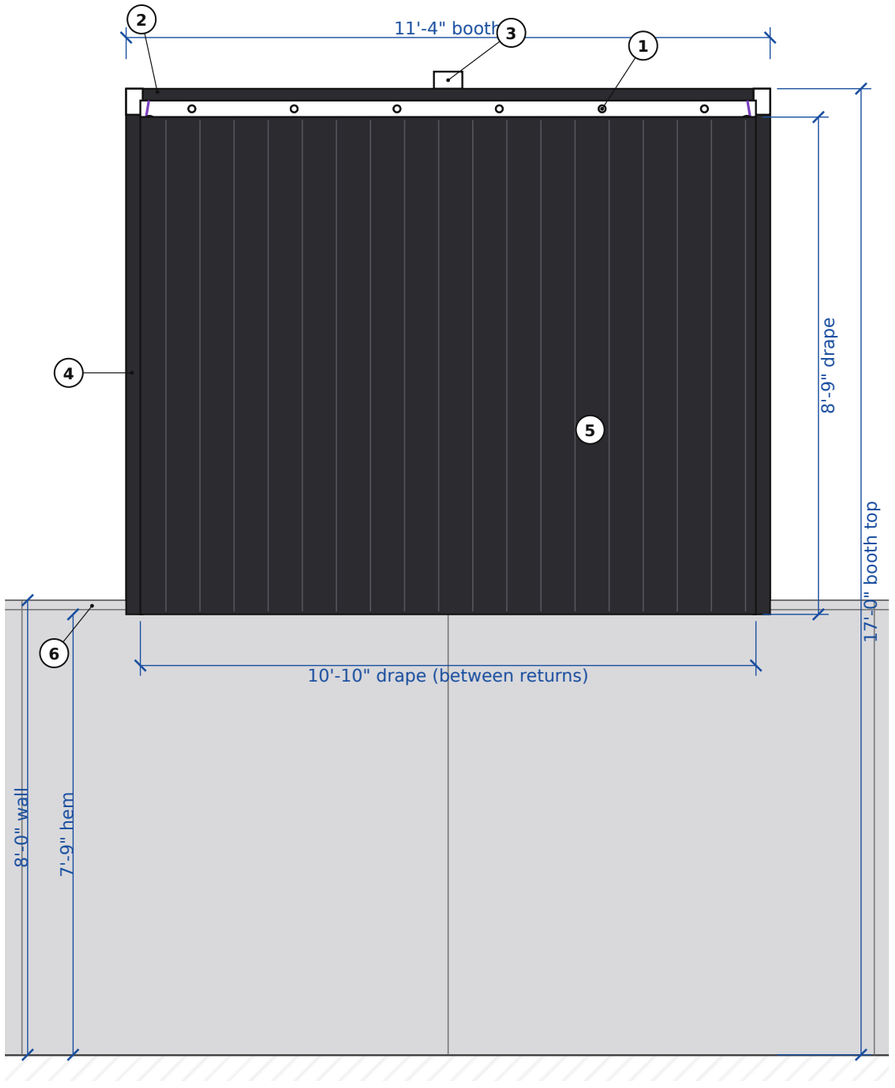
KEYED NOTES

1. Kabuki drape: IFR black velour, chain in the hem pocket.
2. 2x4 batten, flown on two lift lines, 6 kabuki pins (D3, D4).
3. 2x6 outrigger, bolted to the booth corner post, pulley at end.
4. Return (beyond): 2x4 frame, black, closes the drape's sides.
5. Booth front lower wall: 2x4 @ 24" o.c. Beveled double top plate carries the deck's low end.
6. Maze wall (existing). Carries no rig load.
7. Lip: two layers 3/4" ply, bullnose edge, carpet wrapped.
8. Deck: 3/4" ply on 2x4 joists @ 16" o.c., down the slope.
9. Beam: (2) 2x6 on 4x4 posts, top beveled 12 deg.
10. 4x4 ramp post, two per beam, 7'-6" apart (A-4).
11. Guide rail (beyond): 3/4" ply x 18", padded 2x4 cap.
12. Stop bar: 2x4 on edge, hinged, gravity latches (D1, D2).
13. Back stop: 2x4 on edge. Holds the ball while reloading.
14. Hoist pulley at 14'-6" on the back wall (reload).
15. Operator platform: 3/4" ply on 2x6, 4x4 legs, top 5'-0".
16. Booth back wall: 2x4 @ 24" o.c., built in two lifts.
17. Guardrail 42" with mid rail. T-handle and cleat rail on it.

Notes

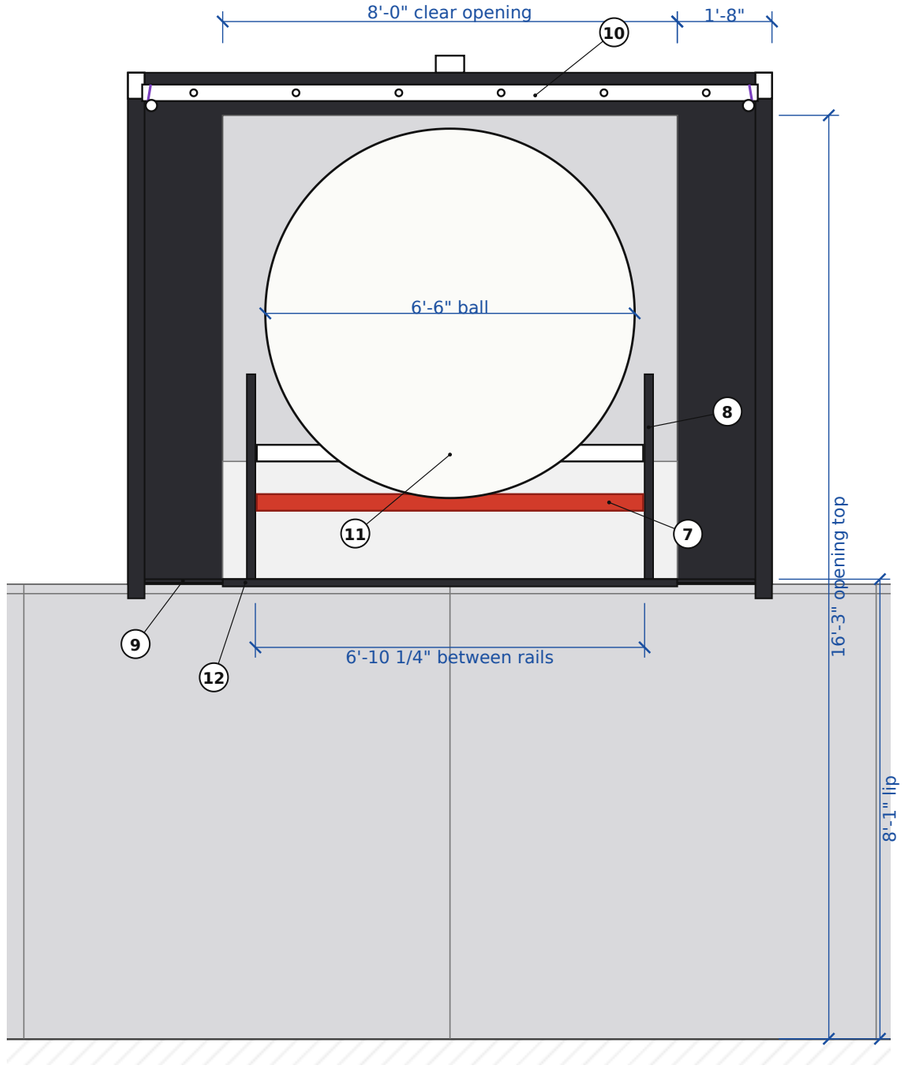
1. The booth stands on the floor behind the maze wall, its front face against the wall's back face. Nothing bears on the wall.
2. The lip layer clears the wall cap by 2 1/8" at the lane face. Joists stop at the booth front wall; only the 1'-6" doubled ply lip cantilevers over the wall.
3. Heights are to the deck surface. Frame the deck, then cut posts to suit and scribe to the floor.
4. Snap the bar line on the ply, measured along the deck from the lip (dimension above the deck).
5. The ball clears the jambs by 9" each side and the header by about 1'-5" as it rolls out.
6. Paint everything a visitor could see flat black.

GYM HAUNTED HOUSE: HIDDEN BALL DROP (GROUP 4)	SHEET
Section A-A through the ramp Scale: 3/8 in = 1 ft (on a 17 x 11 in sheet) Planning sketch: verify every dimension on site before cutting. From the sandbox model (src/scenes/ball-drop-rig.js), scripts/drawings.mjs	**A-2** 2026-10-17



ELEVATION B: SHOW CONDITION

Seen from the group 4 lane, drape hung and armed. Only black surfaces show.



ELEVATION C: DRAPE REMOVED

Same view with the drape and batten left off to show the opening and ramp.

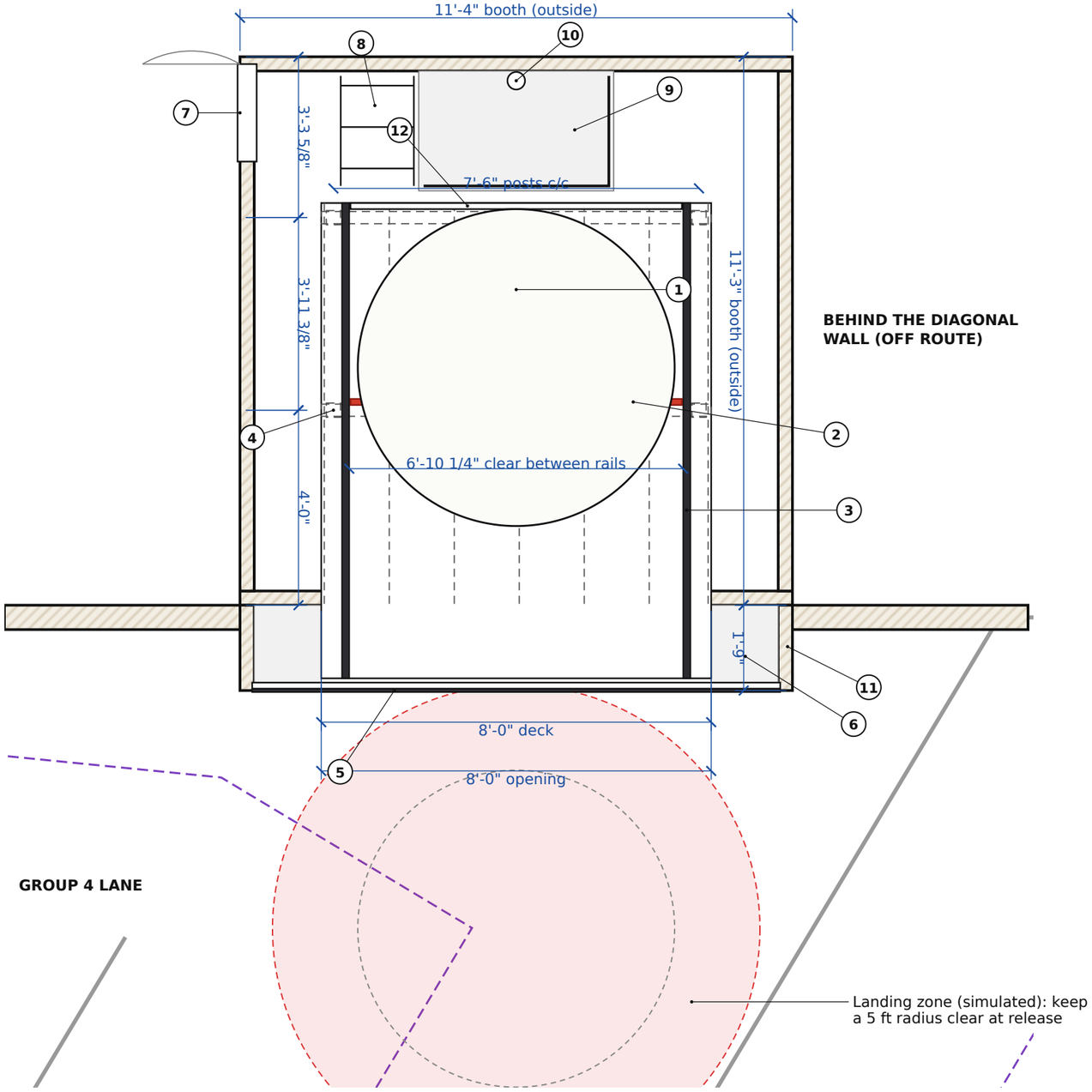
KEYED NOTES

1. Kabuki pins (6) on the batten (A-5, D3 and D4).
2. Lift line to the outrigger pulley, leads back to a cleat in the booth.
3. Lane camera on the header, for the operator's monitor.
4. Return, seen end-on.
5. Drape: IFR black velour.
6. Maze wall (existing), 8'-0".
7. Stop bar, 2x4 on edge, 6'-9 5/8".
8. Rail: 3/4" ply x 18", padded cap.
9. Lip filler: 3/4" ply, black.
10. Header: 2x4 frame, black sheeting.
11. Back stop (reload).
12. Lip: doubled ply, carpet wrapped.

Notes

1. Booth front, jambs, and header are framed like the maze walls and covered in the same black sheeting.
2. The drape fills the width between the returns. With the returns, the lip fillers, and the hem below the lip, it closes every sight line from the lanes (checked in the model from every route position at 3'-6", 5'-6", 6'-6" eyes).
3. Returns run from the hem to the booth top, notched over the maze wall cap.
4. Ball at rest: top at 16'-0", 5 ft back from the opening. It only rolls downhill, so it never nears the header.

GYM HAUNTED HOUSE: HIDDEN BALL DROP (GROUP 4)	SHEET
Elevations from the lane Scale: 3/8 in = 1 ft (on a 17 x 11 in sheet) Planning sketch: verify every dimension on site before cutting. From the sandbox model (src/scenes/ball-drop-rig.js), scripts/drawings.mjs	**A-3** 2026-10-17



PLAN (TOP VIEW, BOOTH ROOF OPEN)

Square to the diagonal wall. Lane at the bottom. Structure under the deck dashed.

KEYED NOTES

1. 6'-6" ball at rest against the stop bar.
2. Stop bar (A-5, D1 and D2).
3. Guide rail, 3/4" ply x 18".
4. 4x4 ramp post (4). Beams dashed under the deck.
5. Drape on the batten, 1'-1 3/4" in front of the wall's lane face.
6. Lip filler (2), at lip height.
7. Door 2'-0" x 6'-8" in the left side wall, opens out.
8. Ladder to the platform.
9. Platform 4'-0" x 2'-5 1/2", guardrail on open sides.
10. Hoist pulley on the back wall at 14'-6".
11. Return (2), continues the booth side out over the lane.
12. Back stop at the deck's high end.

Notes

1. Set out the booth from the opening center on the wall (30'-10 7/8" from the diagonal wall's left end, sheet A-1). Snap a centerline square to the wall through that point.
2. The booth is a box on its own: front, two sides, and back wall frames, 2x4 at 24" o.c., corner posts tied together at the top. Brace and anchor it (A-7, A-8): it is 17 ft tall.
3. Joists run down the slope at 16" o.c. Two beams on 4x4 posts and the booth front wall carry them.
4. The left side door opens out; keep the path to it clear.
5. The platform sits against the back wall, behind the deck's high end, so the operator's head is clear of the deck.
6. The landing zone and route are from the layout model; the dashed purple line is the visitor route.

SET-OUT POINTS (gym plan coordinates, ft)

Point	x	y
Opening center on wall	x 46.50	y 20.10
Booth front, left corner	x 41.51	y 22.80
Booth front, right corner	x 51.23	y 16.97
Booth back, left corner	x 35.72	y 13.15
Booth back, right corner	x 45.44	y 7.32
Ramp lip, center	x 47.14	y 21.17
Rear posts, left / right	x 39.07	y 15.00
	x 45.50	y 11.14
Mid posts, left / right	x 41.10	y 18.39
	x 47.53	y 14.53
Landing spot (simulated)	x 49.79	y 25.58

GYM HAUNTED HOUSE: HIDDEN BALL DROP (GROUP 4)

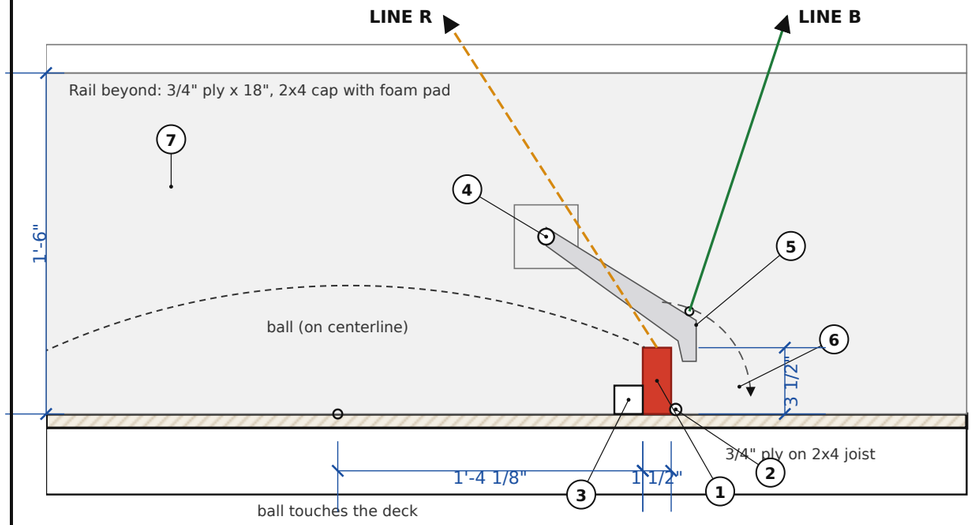
SHEET

Plan of the rig

Scale: 3/8 in = 1 ft (on a 17 x 11 in sheet)
Planning sketch: verify every dimension on site before cutting.
From the sandbox model (src/scenes/ball-drop-rig.js), scripts/drawings.mjs

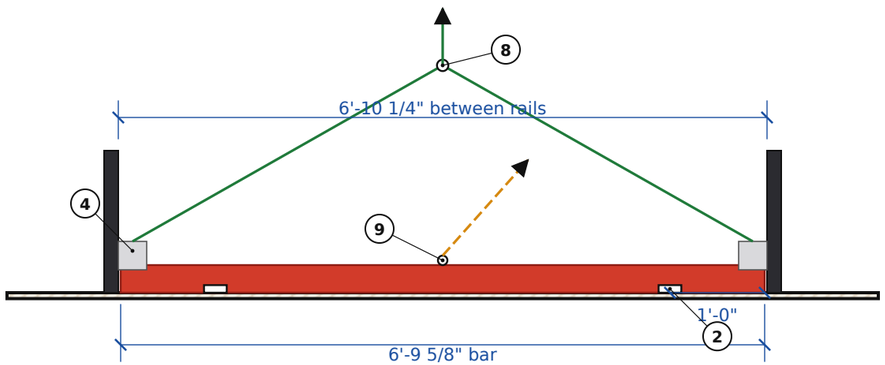
A-4

2026-10-17



D1 STOP BAR AND GRAVITY LATCH (SECTION)

1 1/2 in = 1 ft. Drawn square to the deck (deck shown level). Lane to the right.

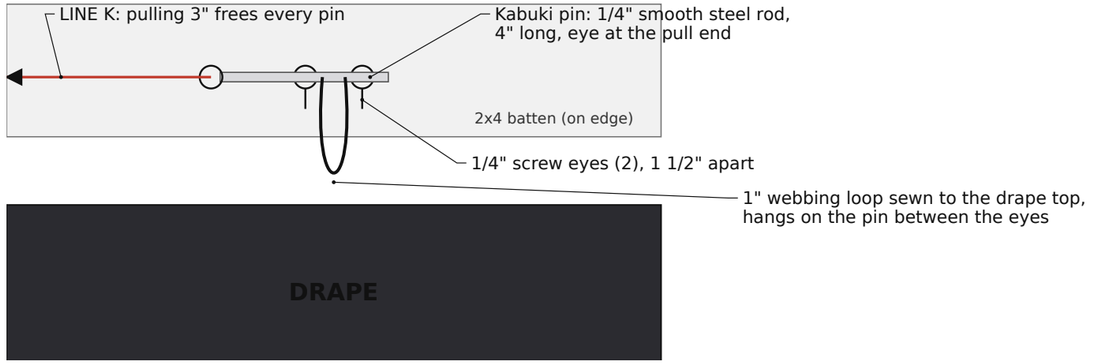


D2 STOP BAR FROM BEHIND (LOOKING DOWNHILL)

5/8 in = 1 ft. Both latch arms tie to one bridle on line B; line R ties to a ring at the bar center.

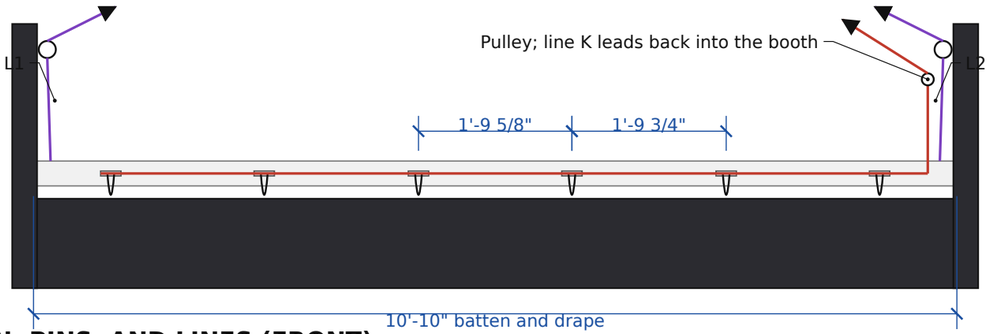
KEYED NOTES (D1, D2)

1. Stop bar: 2x4 on edge, 6'-9 5/8", painted red so it is easy to check.
2. 3" butt hinges (2), lane-side bottom edge, 1'-0" from each end: folds toward the lane.
3. Over-travel block, 1 1/2" sq: stops the bar going past upright when it is reset.
4. Gravity gate latch on each rail's inner face, about 1'-6" clear of the ball at this height.
5. Hooked tip drops in front of the bar top; its sloped back lets the bar push past on reset.
6. On release the ball knocks the bar flat and rolls over 1 1/2" of bar.
7. Rail beyond: 3/4" ply x 18", 2x4 cap with a foam pad.
8. Bridle ring: both latch arms to line B and the T-handle.
9. Ring at the bar center for line R (reset).



D3 KABUKI PIN STATION (FRONT)

3 in = 1 ft. One of 6 stations along the batten.



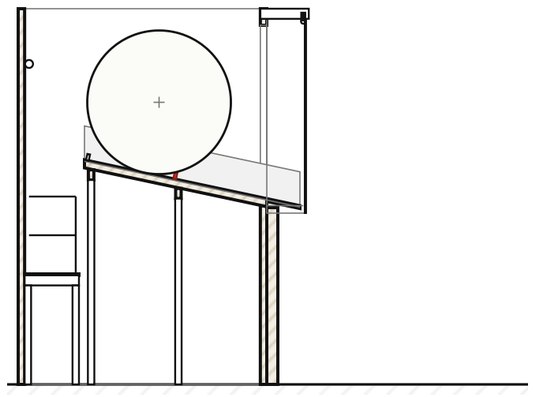
D4 BATTEN, PINS, AND LINES (FRONT)

9/16 in = 1 ft. Line K runs along all 6 pins and out over the right outrigger; L1 and L2 fly the batten.

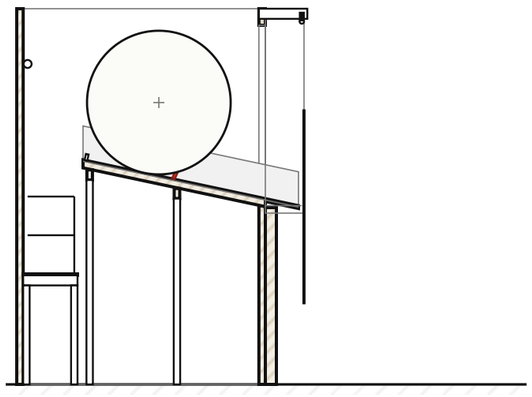
HOW THE TRIGGER WORKS

1. Armed: the bar stands upright between the rails, held by two gravity latches. The ball rests against it and pushes with only about 2 1/2 lb (12 lb ball on a 12 deg slope), so the latches and the pull see very little load.
2. The drape hangs on 6 webbing loops, one over each kabuki pin. Every pin points the same way along the batten and is tied to line K.
3. The operator pulls one T-handle (A-6). Line K moves 3" and all 6 pins slide out of their loops together: the drape falls.
Line B has 2" of extra slack, so it lifts the latch arms a moment later: the bar is free, the ball knocks it flat and rolls.
4. The drape is heaped on the lane floor in about 1 s; the ball reaches the drape line at about 2.6 s.
5. Reset: haul the ball up past the bar (line H), pull line R to stand the bar up (the latches click over it), ease the ball down onto the bar, trip the snap shackle, re-hang the drape on the lowered batten, fly it out, and set line K's slack. See A-8.

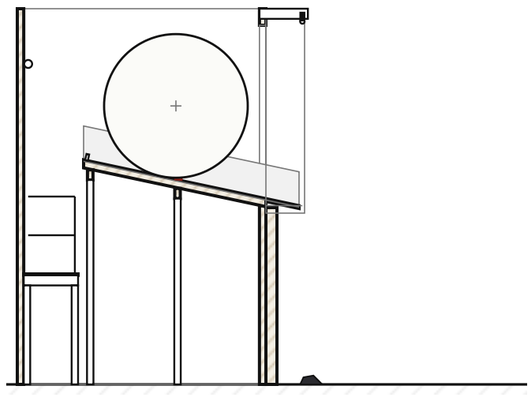
GYM HAUNTED HOUSE: HIDDEN BALL DROP (GROUP 4)	SHEET
Mechanism details: stop bar and kabuki drop Scale: as noted (on a 17 x 11 in sheet) Planning sketch: verify every dimension on site before cutting. From the sandbox model (src/scenes/ball-drop-rig.js), scripts/drawings.mjs	**A-5** 2026-10-17



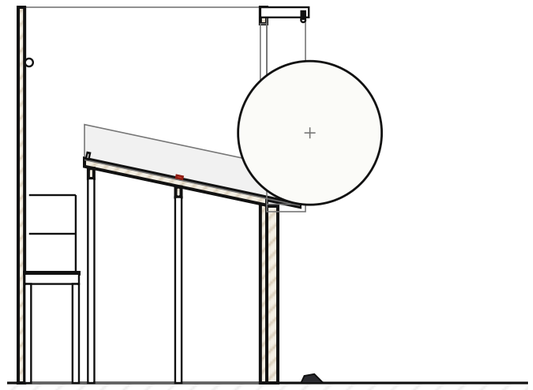
t = 0.0 s
ARMED. Ball on the bar, drape hung.



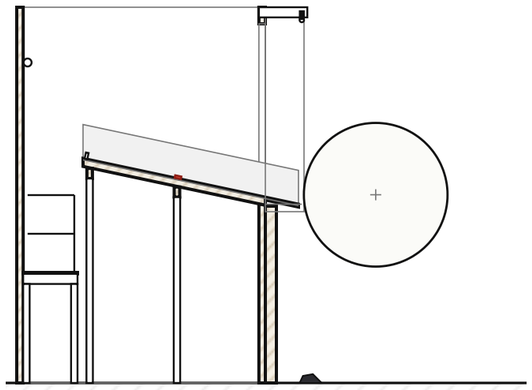
t = 0.5 s
Pull. Drape falling, bar tipping.



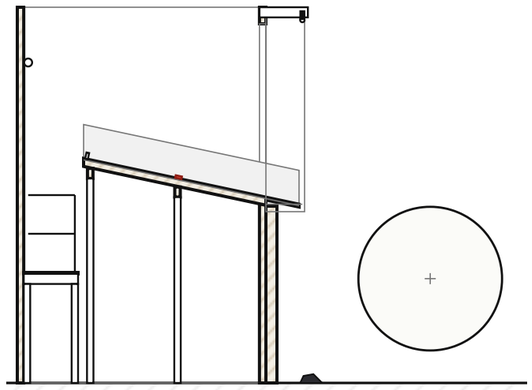
t = 1.2 s
Drape down, bar flat. Ball rolling.



t = 2.6 s
Ball reaches the lip.

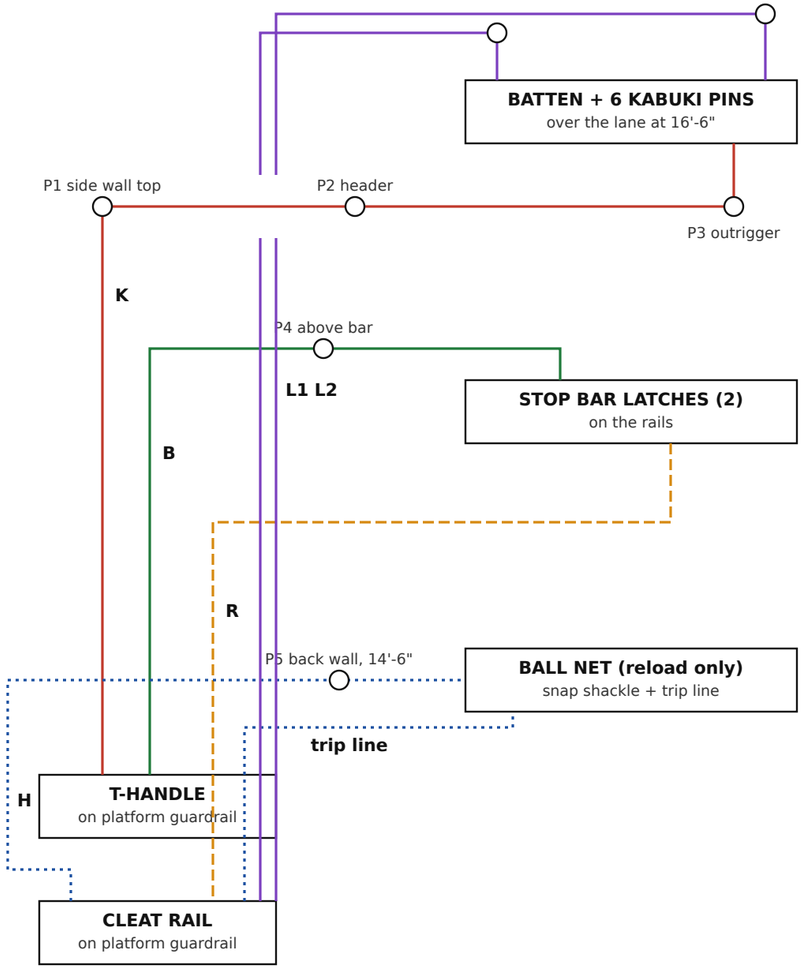


t = 3.1 s
Over the wall, dropping.



t = 3.6 s
Landed. Rolls on down the lane after the group.

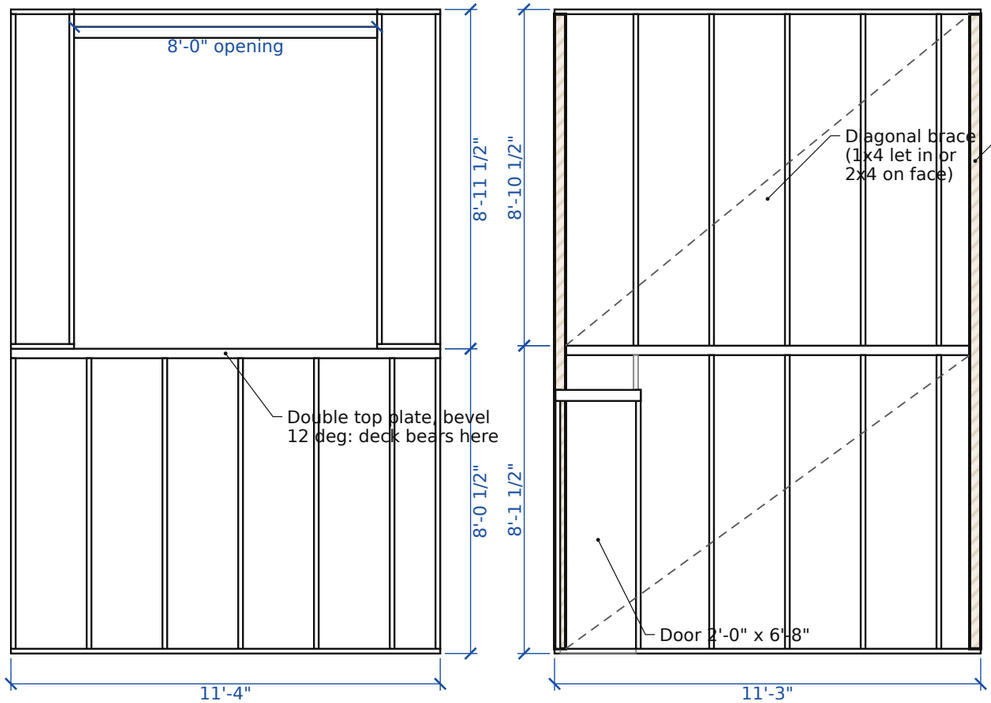
RIGGING (NOT TO SCALE)



- K Kabuki release: handle to all 6 pins (taut).
- B Bar release: handle to both latch arms, 2" slack.
- R Bar reset: bar ring to cleat (slack in show).
- H Hoist + trip line: reload only, never in show.
- L1, L2 Batten lift lines: fly out to stops, cleat.

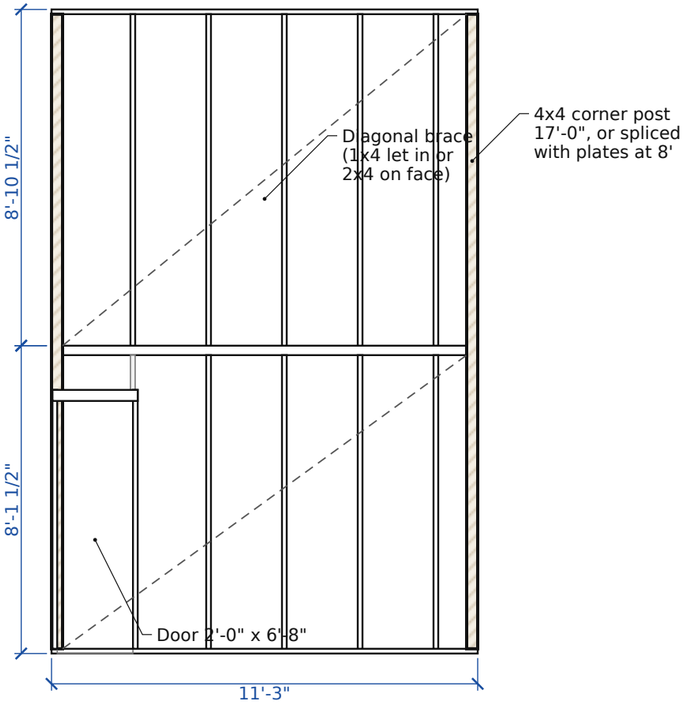
1/4" braided polyester line, black. Pulleys: 1 1/2" single blocks on screw eyes or eye bolts. Handle travel about 6": K frees the drape in the first 3", B frees the bar after 5". Test the pull 20 times before any show (A-8).

GYM HAUNTED HOUSE: HIDDEN BALL DROP (GROUP 4)	SHEET
Rigging and the release sequence Scale: section frames 3/16 in = 1 ft; rigging not to scale (on a 17 x 11 in sheet) Planning sketch: verify every dimension on site before cutting. From the sandbox model (src/scenes/ball-drop-rig.js), scripts/drawings.mjs	**A-6** 2026-10-17



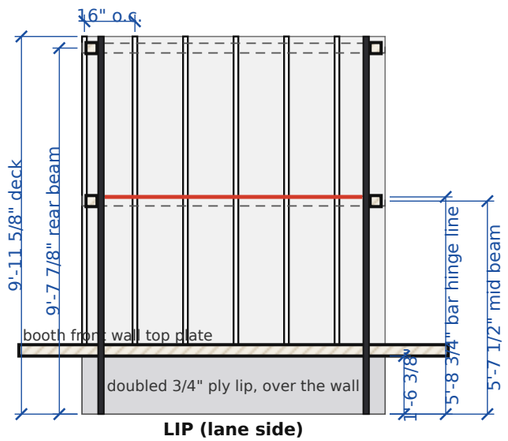
FRONT WALL FRAMING

Seen from inside the booth. Built in two lifts.



LEFT SIDE WALL FRAMING

Seen from outside. Right side the same, no door.



LIP (lane side)

DECK FRAMING PLAN (PLYWOOD OFF)

1/4 in = 1 ft. Distances measured along the slope from the lip. Lane at the bottom.

CUT LIST (ESTIMATE, CHECK ON SITE)

Part	Material	Qty	Size / length
Booth front, lower wall	2x4	8 + 3	studs 7'-8"; plates 11'-4", top two beveled 12 deg
Booth front, jambs + header	2x4	8 + 6	studs 8'-8 1/2"; plates and header 8'-0"
Booth sides (each)	2x4	16 + 4	studs 7 @ 7'-9" + 7 @ 8'-9", door; plates 11'-3"
Booth back	2x4	16 + 4	studs 7 @ 7'-9" + 7 @ 8'-9", blocking; plates 11'-4"
Corner posts	4x4	4	17'-0" (or 8' + 9' spliced)
Bracing	1x4 or 2x4	6	diagonals, each wall, both lifts
Booth sheathing	black sheeting	~700 sf	match the maze walls
Deck joists	2x4	7	8'-5 1/4" along slope, 16" o.c.
Deck beams	2x6	4	8'-0" (doubled, 2 beams)
Ramp posts	4x4	4	mid 8'-4 7/8", rear 9'-3"
Deck top	3/4" plywood	3 sh	9'-11 5/8" x 8'-0"
Lip doubler	3/4" plywood	1	1'-6 3/8" x 8'-0", bullnose edge
Guide rails	3/4" plywood	2	18" x 9'-11 5/8" (splice backed)
Rail caps	2x4 + foam	2	9'-11 5/8", pipe insulation pad
Stop bar / back stop	2x4	2	6'-9 5/8"
Over-travel blocks	2x2	2	6"
Lip fillers	3/4" plywood	2	1'-4 1/2" x 1'-7 3/4"
Returns	2x4 + sheeting	2	1'-9" x 9'-3", notch at wall cap
Outriggers	2x6	2	3'-2 1/4", bolted to corner posts
Batten	2x4 (straight)	1	10'-10"
Platform	3/4" ply, 2x6, 4x4	1	4'-0" x 2'-5 1/2", top 5'-0"
Guardrail	2x4	6	42" posts, top + mid rails
Kabuki drape	IFR black velour	1	10'-10" x 8'-9" finished, 6 loops, hem chain

HARDWARE

Item	Qty	Notes
Gravity gate latch (steel)	2	stop bar, on rail inner faces
3" butt hinges	2	stop bar to deck, lane-side edge
1/4" screw eyes	14	12 for kabuki pins, spares
1/4" smooth steel rod (kabuki pins)	6	4" long, one end bent to an eye
1 1/2" single pulley blocks	8	K, B, H, L1, L2 routing
1/4" braided polyester line, black	200 ft	all lines
Cleats	4	cleat rail on the platform guardrail
T-handle	1	pull handle, K and B tied off
Snap shackle with trip line	1	hoist line to ball net (reload)
Carpet	8 x 2 ft	wrap the lip
Lane camera + monitor	1	header camera, monitor on platform
Anchors / ballast	as req.	booth to floor or to framing, per owner

GYM HAUNTED HOUSE: HIDDEN BALL DROP (GROUP 4)

SHEET

Framing, cut list, and hardware

Scale: 1/4 in = 1 ft (framing elevations) (on a 17 x 11 in sheet)
Planning sketch: verify every dimension on site before cutting.
From the sandbox model (src/scenes/ball-drop-rig.js), scripts/drawings.mjs

A-7

2026-10-17

ARMING (BEFORE EACH GROUP)
1. Ball rests against the upright stop bar. Both latches down.
2. Drape hung on all 6 pins, batten flown to its stops, L1/L2 cleated.
3. Line K snug, line B with its 2" of slack, line R slack.
4. Hoist line H unclipped and pulled back into the booth.
5. Drape heap cleared from the lane. Lane clear.
6. Radio check with the lane wrangler: "Armed."

RELEASE RULE
Pull only when ALL of these are true on the monitor:
- the group has walked well past the landing spot (about 10 ft past it, on toward the stage end);
- nobody is entering from group 3 (the doorway is just a few feet from the landing spot);
- no staff member is in the landing zone.
If in doubt, do not pull. Skipping one group is fine.

THE PULL
One smooth pull on the T-handle, about 6".
Call "Ball out." The wrangler steers the ball and keeps the group moving toward the exit.

TIMING (SIMULATED)
Drape on the floor: about 1 s.
Ball leaves the lip: 2.6 s. Lands: 3.5 s, at about 13 mph.
Rolls down the lane at about walking pace and stops at the stage end within about a minute.

RESET (BETWEEN GROUPS)
1. Wrangler walks the ball back up the lane to the wall under the opening.
2. Operator lowers line H from the hoist pulley; the wrangler clips the snap shackle to the ball's net.
3. Operator hauls the ball up the wall face, over the lip, and up the ramp past the flat stop bar to the back stop. Cleat line H.
4. Pull line R: the bar stands up and the latches click over it. Check both latch arms are down.
5. Ease line H until the ball rests on the bar. Pull the trip line to open the snap shackle; pull line H back.
6. Lower the batten on L1/L2 to about 6 ft over the lane. The wrangler picks up the drape and hooks its 6 loops on the pins (each pin pushed through both eyes and its loop).
7. Fly the batten back to its stops, cleat L1/L2.
8. Take up line K so the pins are snug; set line B's 2" slack.
9. Clear the lane, then arm (left column).

Time the reset in rehearsal; it must fit the gap between groups. A second ball and drape would halve it.

SAFETY
- Landing zone: the ball lands at about 13 mph on the route's turn corner. The release rule is the main safeguard.
- Wrangler in the lower lane, in black, on the radio.
- Booth: 17 ft tall on a small footprint. Brace all four walls, tie the corner posts at the top, and anchor or ballast the base. Agree the method with the maze owner.
- No rig load on the maze wall.
- Platform guardrail 42" with mid rail on open sides.
- Drape and booth fabric: inherently flame-retardant (NFPA 701). Ask the fire marshal early.
- No exposed screws or staples where the ball touches: deck, rails, lip, booth edges. Carpet the lip.
- Keep hands clear of the bar when latching it; load the ball only after the bar is latched (step 4 then 5).
- The fallen drape is a trip hazard: clear it before the next group enters.

TEST BEFORE ANY SHOW
1. Pull 20 times without the ball: all pins and both latches release every time, in order (drape, then bar).
2. 10 releases with the ball and nobody in the lane. Mark the landing spot; compare with A-1 (within about 1 1/2 ft).
3. Walk the route at kid and adult eye height with show lighting: nothing of the rig shows.
4. Rehearse the reset with the real crew and time it.

GYM HAUNTED HOUSE: HIDDEN BALL DROP (GROUP 4)	SHEET
Operating, reset, and safety Scale: none (on a 17 x 11 in sheet) Planning sketch: verify every dimension on site before cutting. From the sandbox model (src/scenes/ball-drop-rig.js), scripts/drawings.mjs	**A-8** 2026-10-17